Operating system and system programming

Linux System Diagnostics Toolkit

2510080029-Sudeeptha
2510080037-Latika
2510080043-Sai Siri
2510080061-Kovidha

UNDER THE GUIDANCE OF DR.KALPANA

PROBLEM STATEMENT

- Linux systems provide many diagnostic commands (such as top, df, free, ps, and ifconfig), but users must execute each command separately to analyze system performance.
- This process is time-consuming and inconvenient, especially for beginners and administrators who need a quick overview of system health.
- Our project aims to develop a Linux System Diagnostics Toolkit that integrates essential system monitoring features into a single, easy-to-use application, providing comprehensive diagnostics and report generation.

OBJECTIVES

- Develop a unified Linux diagnostics toolkit.
- Display system information and running processes.
- Generate detailed diagnostic reports.
- Provide a simple command-line interface.
- Improve troubleshooting efficiency for Linux users.

TECHNOLOGY STACK

Linux System Diagnostics Toolkit

OPERATING SYSTEM

ubuntu

Ubuntu Linux
(22.04 LTS or later)

PROGRAMMING LANGUAGE

python™

Python 3
(3.10 or later)

LIBRARIES

psutil
platform
socket
os
subprocess

Core Python libraries for system monitoring and diagnostics

k

VERSION CONTROL

git

Git & GitHub

IDE / CODE EDITOR

Visual Studio Code

Code editing, debugging and project management

REPORT FORMAT

TXT

Text File
(.txt)

ADDITIONAL TOOLS

Linux Terminal (CLI)

For system commands and execution

All technologies used are open-source and cross-platform compatible.

EXPECTED OUTCOMES

The Toolkit will provide:

- Complete system information
- CPU utilization monitoring
- Memory and swap usage
- Disk usage analysis
- Network diagnostics
- Running process monitoring
- Automatic system health report generation
- A user-friendly command-line interface

Benifits:

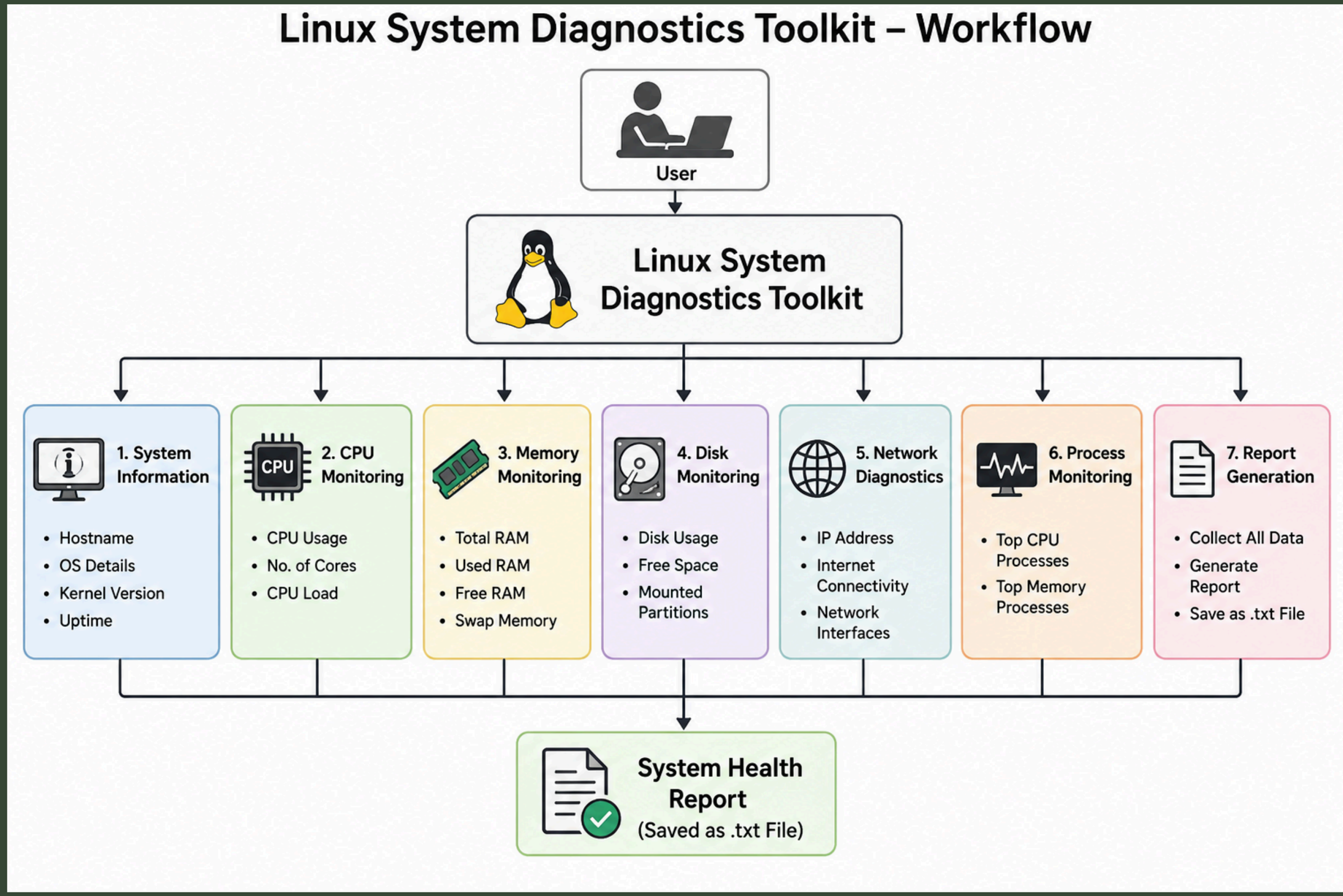
- Faster troubleshooting
- Easy monitoring
- Improved system maintenance
- Suitable for students, developers, and system administrators

PROJECT MODULES

Module 1:
System Information
Module 2:
CPU Monitoring
Module 3:
Memory Monitoring
Module 4:
Disk Monitoring

Module 5:
Network Diagnostics
Module 6:
Process Monitoring
Module 7:
Report Generation

Linux System Diagnostics Toolkit – Workflow



Project Plan (Timeline)

week	Activity
Week 1	Requirement Analysis & Project Setup
Week 2	System Information Module
Week 3	CPU and memory monitoring
Week 4	Disk Monitoring
Week 5	Network Diagnostics
week 6	Process Monitoring
week 7	Report Generation
week 8	Testing, Bug Fixing & Documentation

Thank You